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1 Introduction

Quantum computation and sensing has the potential to revolutionize industries from cryptography to medicine. Right now, the effectiveness of these systems are severely handicapped by the presence of persistent, inherent noise. We propose a physics-informed, transformer-based denoising autoencoder for quantum state noise mitigation. We believe that the properties of non-locality inherent to transformers offer a superior reflection of the non-local nature of quantum entanglement, compared to the inherently local perspective imposed by traditional CNN architectures.

2 Methods

2.1 Quantum Circuit Dataset Generation and Density Matrix Extraction

In order to aid supervised learning, we fabricate a large scale, synthetic dataset with known "ground truth." We use a quantum programming library such as Cirq to generate 10,000 random 5-qubit random quantum circuits. The circuits are implemented in a controlled manner such that we cover different noise regimes: there are 500 instances for each combination of five types of noises, and four levels of intensity (5 **4** 500 = 10,000 circuits). In addition, we use an 80/20 train-test split prior to training to ensure unbiased testing.

2.2 Noise Modelling

The resulting noisy circuits are simulated with Cirq's DensityMatrixSimulator, and are the inputs that will be given to the network. The types of noise applied are as followed:

- **Bitflip:** Introduces random Pauli-X operations on qubits, flipping the computational basis state [[file:media/image5.png]] with a given probability.
- **Depolarizing:** With a given probability, replaces the qubit's state by the maximally mixed state, reducing its purity uniformly across all bases.
- **Amplitude damping:** Models energy relaxation, where the excited state [[file:media/image4.png]] decays to the ground state [[file:media/image1.png]], mimicking spontaneous emission or photon loss.
- **Phase damping:** Represents loss of phase coherence between states without any exchange of energy. Populations remain unchanged while off-diagonal terms decay.
- **Mixed:** A composite channel that applies a random combination of the above noise mechanisms with equal likelihood, approximating general environmental decoherence.

2.3 Transformer architecture

We used an autoencoding transformer architecture trained to map noisy density matrices onto the clean ones. The architecture is selected for its Transformer-like advantage in capturing long-range, non-local correlations – a necessary condition to adequately reconstruct entanglement and coherence in mult

i-qubit states. The workflow is as follows: The input density matrix is projected into a sequence of patch tokens. This sequence is transformed within the encoder based Transformer network using self-attention layers to capture global noise patterns. The output is then reconstructed via a lightweight decoder to yield the denoised state ,

2.4 Physics-informed loss

Following the physics-informed neural network (PINN) paradigm

We design the training objective to embed quantum mechanical structure directly into the optimization process. The total loss combines a data-driven reconstruction term, mixing mean-squared error, frobenius fidelity, and physical regularizers that enforce Hermiticity, unit trace, and positive semidefiniteness of the predicted density matrices. This formulation constrains the network outputs to the manifold of valid quantum states, ensuring that denoised predictions remain consistent with the postulates of quantum mechanics.

2.5 Training

The model was trained for 100 epochs with a batch size of 16. The training process employed the Adam optimizer with a learning rate of 3e-4. The training was conducted on an NVIDIA T4 GPU, with one epoch taking approximately 22 seconds. Model checkpoints were saved every epoch, and no learning rate scheduling or early stopping was employed.

The Transformer autoencoder uses a patch embedding size of $2{\times}2$, an embedding dimension of 128, 6 encoder blocks, and 8 attention heads per block. Each block includes a $2\times$ expansion MLP and standard pre-norm residual connections. The decoder consists of a single transposed convolution layer projecting back to a 2-channel density matrix representation.

To improve robustness across heterogeneous noise regimes, we adopoted an **adversarial reweighting scheme** during training. Specifically, losses were grouped by noise intensity levels (0.05, 0.10, 0.15, 0.20), and adaptive group weights were updated via an exponentiated-gradient rule:

media/image2.png

Where

$$L_i$$

is the mean loss for group

$$i$$

and

$$\eta = 0.05$$

controls the adversarial learning rate. This mechanism dynamically up-weights samples from higher noise distributions, encouraging the model to prioritize difficult noise conditions and improving generalization across the full noise spectrum.

3 Analysis

3.1 Performance

After training, the model achieved an average fidelity improvement of **0.133** across all noise conditions. However, this gain was not uniformly distributed. As shown in Figure 1, performance deteriorates sharply at higher noise levels due to inputs that lack discernible quantum structure, resulting in low-information or effectively random reconstructions. The transformer contains over 800,000 parameters, while the dataset contains a relatively small number of circuits in comparison, boasting just 10,000. Transformers typically require substantially larger datasets to learn robust, global noise-correlation structures, and this mismatch likely contributes to the weaker generalization observed at high noise levels compared to convolutional baselines[Kendre, 2025]. Furthermore, with limited data, the **physics-informed regularizers** dominate the loss, biasing the model toward producing physically valid, but overly conservative, density matrices rather than closely reconstructing the true clean states. In effect, this over-regularization can degrade the denoising process itself, yieliding negative fidelity deltas in cases where the network overcorrects physically valid but noisy states media/image3.png

3.2 Scalability

These limitations are largely attributable to the data-model scale mismatch rather than the architecture itself. Transformers are known to exhibit strong data-scaling laws: as both model capacity and dataset size increase, their ability to capture long-range dependencies improves predictably[Kaplan et al., 2020; Hoffmann et al. 2022]. In this context, expanding the dataset by several orders of magnitude would allow the network to better model complex, non-local noise correlations that convolutional architectures inherently miss[Vaswani et al., 2017; Dosovitskiy et al., 201]. Furthermore, larger-scale training would rebalance the optimization landscape, reducing the dominance of the physics-informed regularizers and enabling the model to prioritize reconstruction fidelity without sacrificing physical validity[Norambeuna et al., 2023; Raissi et al., 2019]. This suggests that the transformer framework remains a promising foundation for quantum-state denoising, provided it is trained in a sufficiently data-rich regime.

4 Related Implementations

max width=

Feature	Machine learning for Quantum Noise Reduction [Kendre, 2025]	Phys
Core Architecture	Convolutional Neural Network (CNN) autoencoder	Trans
Loss Function	Used a supervised learning approach with a fidelity-aware composite loss function	Uses
Data and Scope	10,000 noisy 5-qubit density matrices	Simil
Key Limitation	CNNs miss non-local entanglement structure	Self-a
Performance	Fidelity = 0.476	Fidel

5 Conclusion

In conclusion, this work introduces the Quantum-Fidelity Physics-Informed Denoising Transformer (QFPIDT), a deep learning framework designed to mitigate quantum noise through a fidelity-aware optimization process. While initial experiments revealed high variance and limited fidelity under mixed noise conditions, these findings highlighted the intrinsic complexity of quantum noise modeling and motivated the transition from convolutional to Transformer-based denoising architectures. Leveraging global self-attention, the Transformer model demonstrates enhanced capacity to restore coherence by capturing non-local dependencies within quantum states.

Future work will involve scaling the training corpus to millions of simulated quantum circuits. Such large-scale data is expected to better exploit the data-hungry nature of attention mechanisms, enabling the Transformer to generalize across diverse noise regimes. Moreover, a proposed follow-up study will compare a standard Transformer with a **physics-informed Transformer**—one constrained by quantum-mechanical priors—to assess how embedding physical laws within the model encourages *truth-seeking* representations over purely statistical correctness. This direction holds promise for developing denoising architectures that bridge empirical learning with fundamental quantum principles, advancing the viability of QFDA in real-world biosensing applications.